This means that $x - x_1 = r(x_2 - x)$

Solving this, $x = \dfrac{rx_2 + x_1}{r+1}$ ..... (1)

We can use this result for points on a line to the general case as follows.

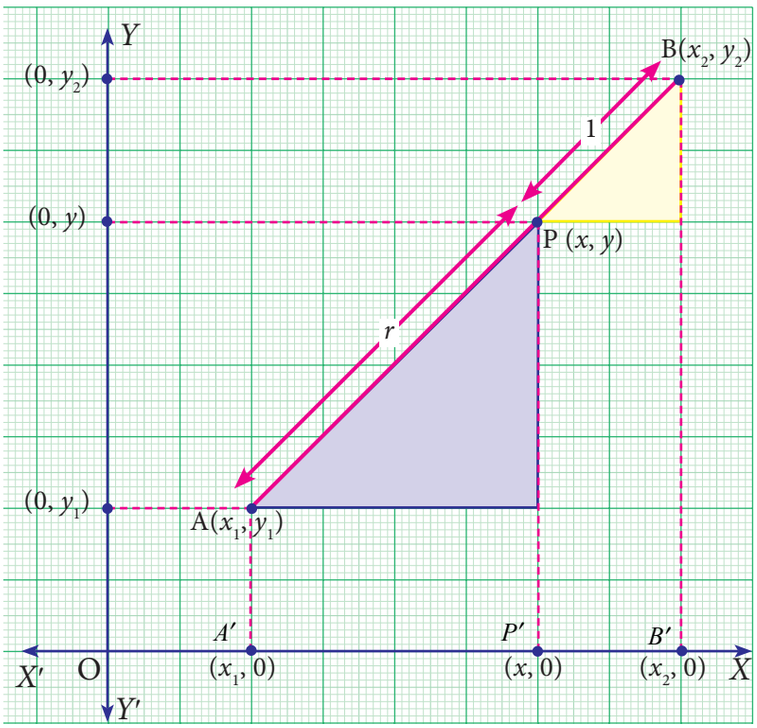


Fig. 5.34

Taking $AP : PB = r : 1$, we get $A'P' : P'B' = r : 1$.

Therefore $A'P' = r(P'B')$

Thus, $(x - x_1) = r(x_2 - x)$

which gives $x = \dfrac{rx_2 + x_1}{r+1}$ ... [see (1)]

Precisely in the same way we can have $y = \dfrac{ry_2 + y_1}{r+1}$

If P is between A and B, and $\dfrac{AP}{PB} = r$, then we have the formula,

P is $\left(\dfrac{rx_2 + x_1}{r+1}, \dfrac{ry_2 + y_1}{r+1} \right)$.

If r is taken as $\dfrac{m}{n}$, then the section formula is $\left(\dfrac{mx_2 + nx_1}{m+n}, \dfrac{my_2 + ny_1}{m+n} \right)$, which is the standard form.

Thinking Corner

(i) What happens when $m = n = 1$? Can you identify it with a result already proved?

(ii) $AP : PB = 1 : 2$ and $AQ : QB = 2:1$.
What is $AP : AB$? What is $AQ : AB$?

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